Parametric vs Observational Salinity: Validation of the SSWD-EvoEpi Seasonal Salinity Model Against WOA23 Climatology

SSWD-EvoEpi Automated Analysis

March 2026

1 Executive Summary

This report evaluates the parametric salinity baseline used in the SSWD-EvoEpi disease model against monthly climatological surface salinity from the World Ocean Atlas 2023 (WOA23) at 0.25° resolution, covering all 896 network nodes from 26.56°N (Baja California) to 61.19°N (western Alaska).

The parametric model assigns each site a constant annual salinity based solely on latitude:

$$S_{\text{ocean}}(\text{lat}) = 31.32 + 0.054 \times (\text{lat} - 50) \tag{1}$$

This formula captures the broad latitudinal gradient with a small overall mean bias of -0.13 psu, but three critical deficiencies emerge:

1. **Zero seasonal variation.** The parametric baseline produces exactly 0.00 psu seasonal range, whereas WOA23 shows a mean range of 1.68 psu across all sites (max 8.11 psu in Prince William Sound). The model entirely misses seasonal freshening that drives disease suppression.

2. **Large regional biases.** The monthly RMSE is 2.42 psu, with errors up to 9.18 psu. The model overestimates salinity by +2.79 psu in the Salish Sea (RMSE = 3.46 psu, $n = 182$) and underestimates by -2.50 psu along the California coast (RMSE = 2.65 psu, $n = 328$). The $R^2 = -0.32$ for annual means, indicating the formula performs worse than simply using the global mean.

3. **Wrong seasonal timing.** The month of minimum salinity varies widely: May (171 sites), June (152), February (116), and August (102). The parametric model's cosine melt pulse peaks on June 15 everywhere, missing the winter-storm freshening in California and late-glacial melt in Alaska.

The disease-modelling consequences are stark. Under the parametric baseline, **zero** of 896 sites experience any June salinity suppression (all values > 30 psu). Under WOA23, **143 sites** show > 1% suppression, with Salish Sea sites averaging 16.0% and reaching 54.5% suppression. The parametric model is blind to this ecologically important signal.

We **recommend replacing the parametric baseline** with WOA23 monthly climatology as the default salinity input.

2 Data Sources & Methods

2.1 WOA23 Surface Salinity

The World Ocean Atlas 2023 provides objectively analysed climatological mean salinity on a 0.25° grid (approximately 20–28 km at these latitudes). We extracted surface salinity (0 m depth) for all 12 calendar months at each of the 896 SSWD-EvoEpi network nodes spanning from 113.15°W to 178.80°W.

Of the 896 sites, 235 (26.2%) fall within a WOA23 ocean grid cell and receive direct extraction. The remaining 661 (73.8%) lie in WOA23 land cells — typically because these coastal and fjord sites sit inside the 0.25° land mask. For these, we used nearest-ocean-cell interpolation. The maximum nearest-neighbour (NN) distance was 0.177° (≈17 km), with a mean of 0.094° (≈9 km), indicating that the nearest ocean cell is typically the immediately adjacent offshore grid cell.

2.2 Parametric Baseline

The current SSWD-EvoEpi model uses Equation 1 to set ocean salinity at each site. At `fw_strength=0` (the baseline), this produces a constant annual value per site with no seasonal cycle. The formula yields values in a narrow band from 30.05 psu (26.56°N) to 31.92 psu (61.19°N). By contrast, the WOA23 data span 22.14–34.31 psu — a 12 psu range compressed by the formula into less than 2 psu.

2.3 Salinity Modifier for Disease Modelling

The *Vibrio* salinity modifier in the SSWD-EvoEpi model uses a nonlinear transfer function:

$$S_{\text{mod}}(S) = \begin{cases} 0 & S \leq 10 \\ \left(\frac{S-10}{28-10}\right)^2 & 10 < S < 28 \\ 1 & S \geq 28 \end{cases} \tag{2}$$

Salinity below 28 psu suppresses *Vibrio* transmission; below 10 psu, transmission ceases entirely. This nonlinear threshold means that even modest salinity differences near 28 psu have disproportionate effects on predicted disease dynamics.

2.4 DFO Lighthouse Climatology

As an independent validation, we compare both models against monthly salinity climatologies from 12 Fisheries and Oceans Canada (DFO) lighthouse stations along the British Columbia coast, representing decades of direct surface salinity measurements at fixed coastal locations.

3 Baseline Comparison

3.1 Annual Mean Scatter

Figure 1 compares parametric and WOA23 annual mean salinity for all 896 sites. While the parametric model clusters tightly between 30.05–31.92 psu, WOA23 annual means span 22.14–34.31 psu. The scatter reveals two dominant bias patterns:

- **Salish Sea / WA** (green, $n = 182$): The parametric model *overestimates* salinity by +2.79 psu on average (RMSE = 3.46 psu). Northern Salish Sea sites (SS-N) show biases up to +5.74 psu because true estuarine salinities (∼26 psu) are far below the parametric prediction.

- **OR / California** (red, $n = 328$): The parametric model *underestimates* by -2.50 psu (RMSE = 2.65 psu). These southern, open-ocean sites have true salinities of 33–34 psu, well above the formula's prediction.
- **British Columbia** (orange, $n = 111$): Best agreement (bias = +0.10 psu, RMSE = 0.61 psu), expected since the formula was calibrated for BC conditions.
- **Alaska** (blue, $n = 275$): Moderate positive bias (+0.67 psu, RMSE = 1.64 psu), with Prince William Sound (AK-PWS) a notable outlier (bias = +2.29 psu) due to heavy glacial freshwater input.

The negative R^2 (-0.32) confirms that the parametric formula does not capture the spatial structure of surface salinity across this domain — it performs worse than simply assigning the global mean to every site.

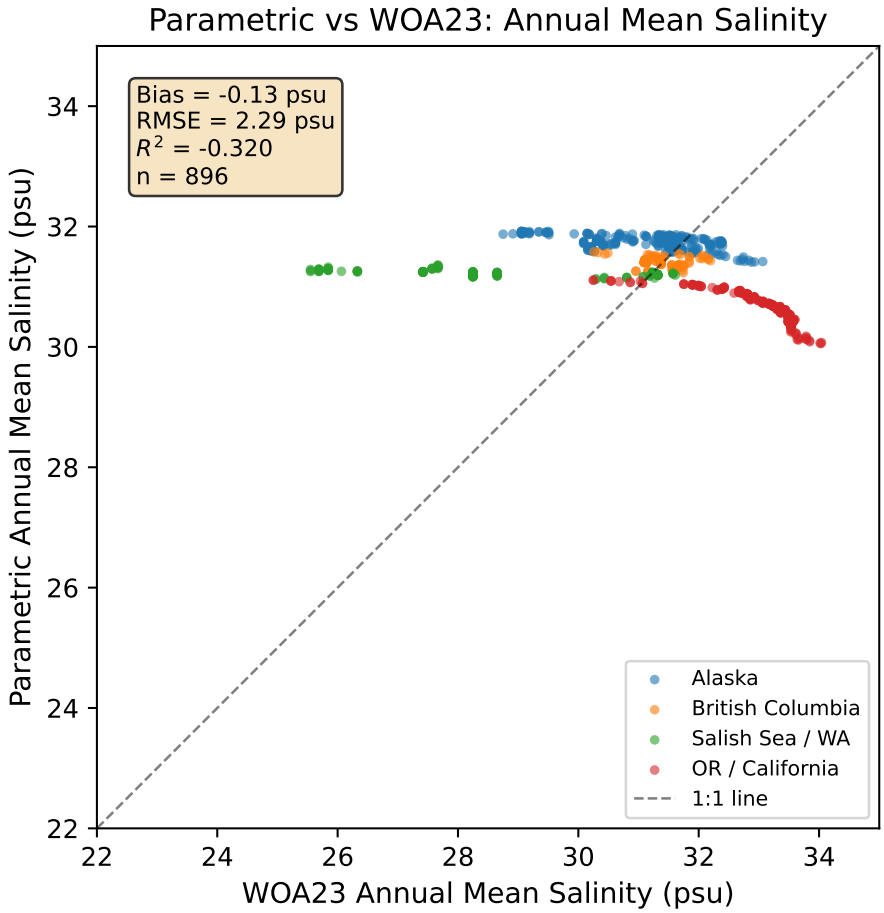

Figure 1: Parametric vs WOA23 annual mean salinity for 896 sites, coloured by region group. Dashed line: 1:1 agreement. The parametric model compresses a 12 psu WOA23 range into a <2 psu band. Bias = -0.13 psu, RMSE = 2.29 psu, $R^2 = -0.32$.

3.2 Geographic Distribution of Bias

Figure 2 maps the annual mean bias geographically. The spatial structure is stark: strong negative bias (blue, parametric too low) dominates California and Baja; strong positive bias (red, parametric too high) marks the Salish Sea and Prince William Sound. The open coast of BC, Washington, and western Alaska shows near-zero bias.

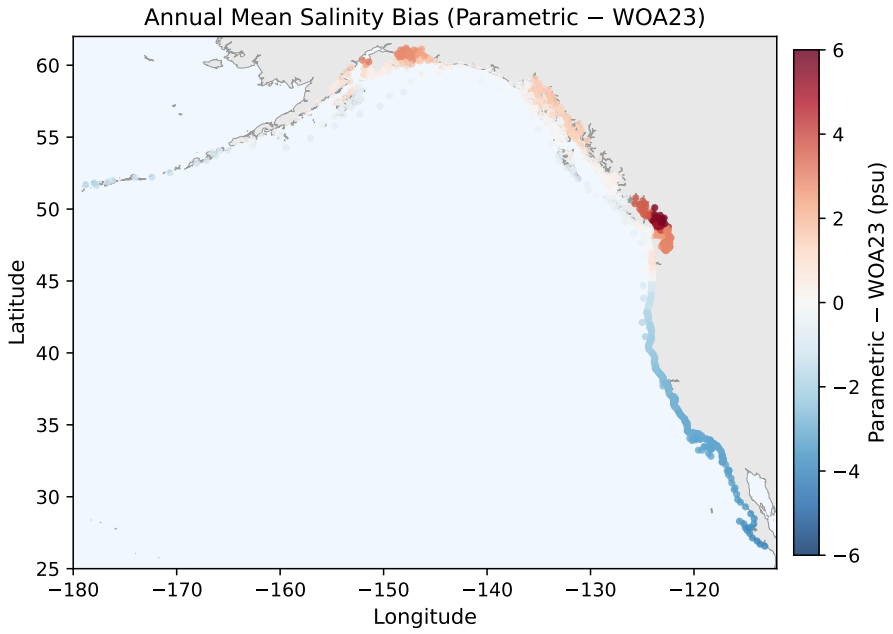

Figure 2: Geographic distribution of annual mean salinity bias (parametric − WOA23). Blue: parametric underestimation; red: overestimation. Scale: ±6 psu.

3.3 Regional RMSE Summary

Table 1 summarises per-region statistics. The worst-performing regions are SS-N (RMSE = 5.14 psu), SS-S (3.60 psu), BJ (3.46 psu), and AK-PWS (3.21 psu). The best are AK-OC (0.37 psu), BC-N (0.53 psu), and WA-O (0.60 psu).

Table 1: Per-region comparison statistics, sorted by RMSE.

Region	n	Bias (psu)	RMSE (psu)	WOA23 $\bar{S}$	Seasonal Range
AK-OC	28	+0.05	0.37	31.66	0.97
BC-N	75	+0.17	0.53	31.28	1.09
WA-O	36	+0.05	0.60	31.14	1.46
AK-EG	22	+0.27	0.72	31.57	1.46
BC-C	36	−0.06	0.73	31.39	0.75
AK-WG	79	+0.17	0.94	31.57	1.22
AK-AL	15	−1.12	1.18	32.59	0.32
AK-FS	48	+0.76	1.25	30.87	2.08
OR	59	−1.05	1.39	32.04	1.35
AK-FN	40	+1.12	1.68	30.62	3.14
CA-N	63	−2.20	2.22	32.98	0.80
JDF	26	+2.37	2.69	28.84	2.11
CA-C	97	−2.79	2.80	33.39	0.38
CA-S	78	−3.10	3.10	33.54	0.19
AK-PWS	43	+2.29	3.21	29.60	6.86
BJ	31	−3.45	3.46	33.67	0.25
SS-S	89	+3.35	3.60	27.87	2.95
SS-N	31	+4.75	5.14	26.55	4.93

4 Seasonal Dynamics

4.1 The Missing Seasonal Cycle

The most fundamental limitation of the parametric baseline is its complete absence of seasonal variation. Every site receives the same salinity in January as in August. In contrast, WOA23 reveals substantial seasonality: the mean seasonal range across all 896 sites is 1.68 psu (median 1.04 psu), with individual sites varying by up to 8.11 psu (Prince William Sound).

This matters for disease modelling because the salinity modifier (Equation 2) is highly non-linear near 28 psu. A site that seasonally drops from 31 psu to 26 psu experiences meaningful transmission suppression that the constant-baseline model cannot capture.

4.2 Timing Heterogeneity

Figure 3 maps the month of minimum WOA23 salinity geographically. If all sites followed a simple summer-freshening pattern, minimum salinity would be concentrated in June–August. Instead, the distribution is heterogeneous:

- **May** — 171 sites (19.1%), the most common minimum month, driven by Alaska and BC sites responding to spring snowmelt.
- **June** — 152 sites (17.0%), peak river discharge in many northern regions.
- **February** — 116 sites (12.9%), dominated by OR/California where winter storms and increased river discharge drive fresher conditions.
- **August** — 102 sites (11.4%), capturing late-summer glacial melt in Prince William Sound and southeastern Alaska fjords.

The geographic map (Figure 3) shows clear spatial coherence: winter minima (blue tones) along California, spring minima (green–yellow) in BC and western Alaska, and summer–autumn minima (orange–red) in the fjords and Prince William Sound. This spatial pattern means the parametric model's single June 15 melt pulse peak is at best a regional average, not a site-appropriate timing.

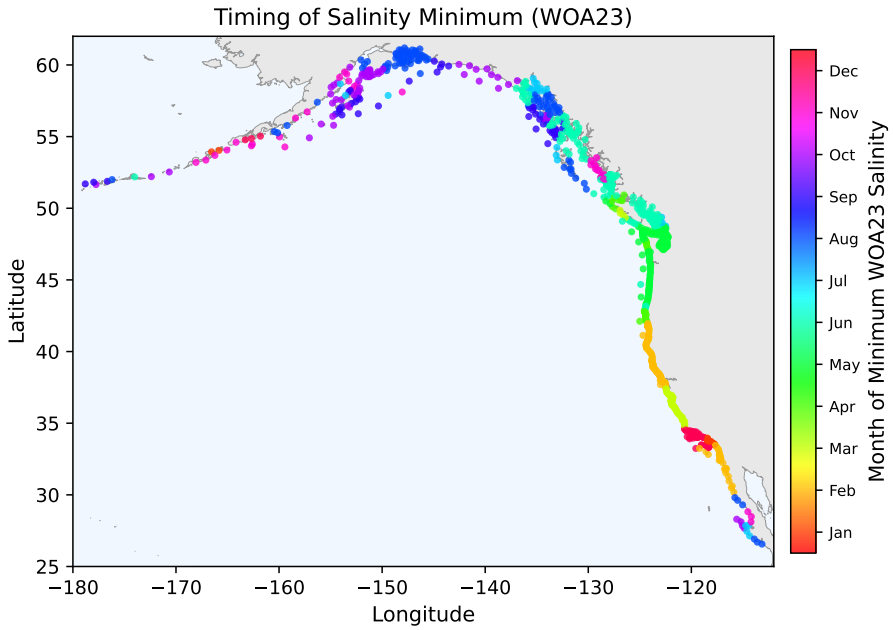

Figure 3: Month of minimum WOA23 salinity for each of the 896 sites. Colour encodes month (cyclic HSV scale). Winter minima dominate California; spring minima dominate BC; summer/autumn minima dominate Alaska fjords. The parametric model assumes a single June 15 peak everywhere.

The stacked histogram in Figure 4 reinforces this pattern quantitatively: salinity minima are distributed across nearly every month, with no single month dominating. The regional colour coding reveals that winter months (Dec–Feb) are dominated by OR/California sites, spring months (Mar–Jun) by Alaska and BC, and late summer (Jul–Sep) by Alaska fjord sites. This heterogeneity is fundamentally incompatible with a single-phase seasonal model.

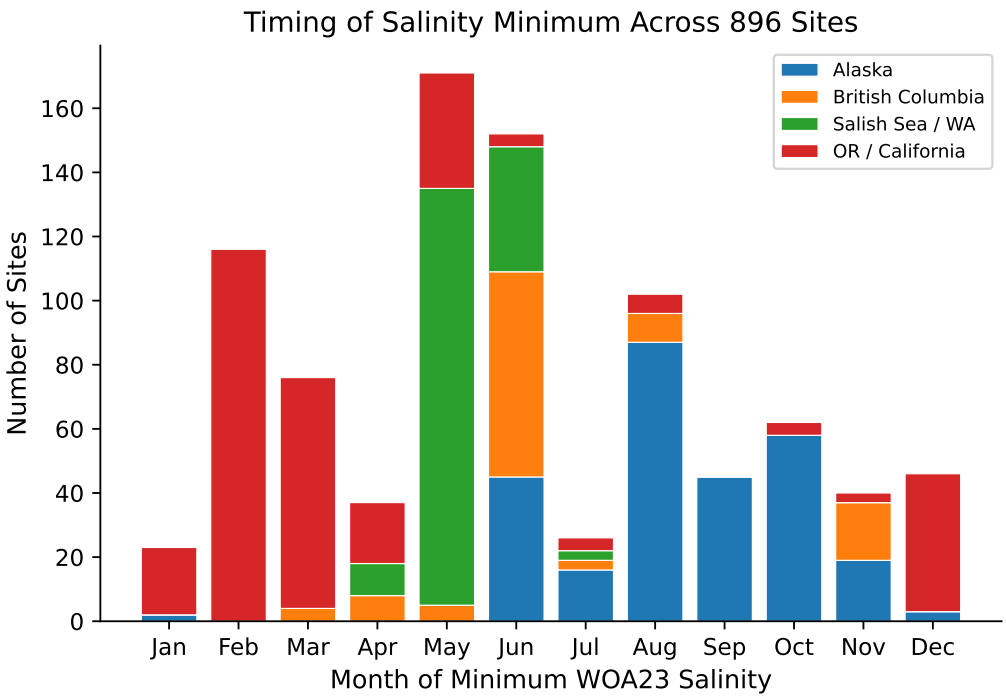

Figure 4: Stacked histogram of the month of minimum WOA23 salinity for 896 sites, coloured by region group. No single month dominates; the four most common months are May (171), June (152), February (116), and August (102).

4.3 Seasonal Range Comparison

Figure 5 directly compares the seasonal range (max − min monthly salinity) between WOA23 and the parametric model. Panel (a) shows the parametric baseline (`fw_strength=0`): every site has exactly zero seasonal range, forming a horizontal line at $y = 0$ while the WOA23 x-axis spans 0–8 psu. Panel (b) shows the parametric model with the melt pulse active (`fw_strength=15`), which introduces seasonality (mean range 3.93 psu) but applies it only at high-latitude, high-enclosedness sites. Many California and Oregon sites retain zero or near-zero parametric range despite WOA23 showing non-trivial seasonal cycles there (0.2–1.4 psu from winter storms).

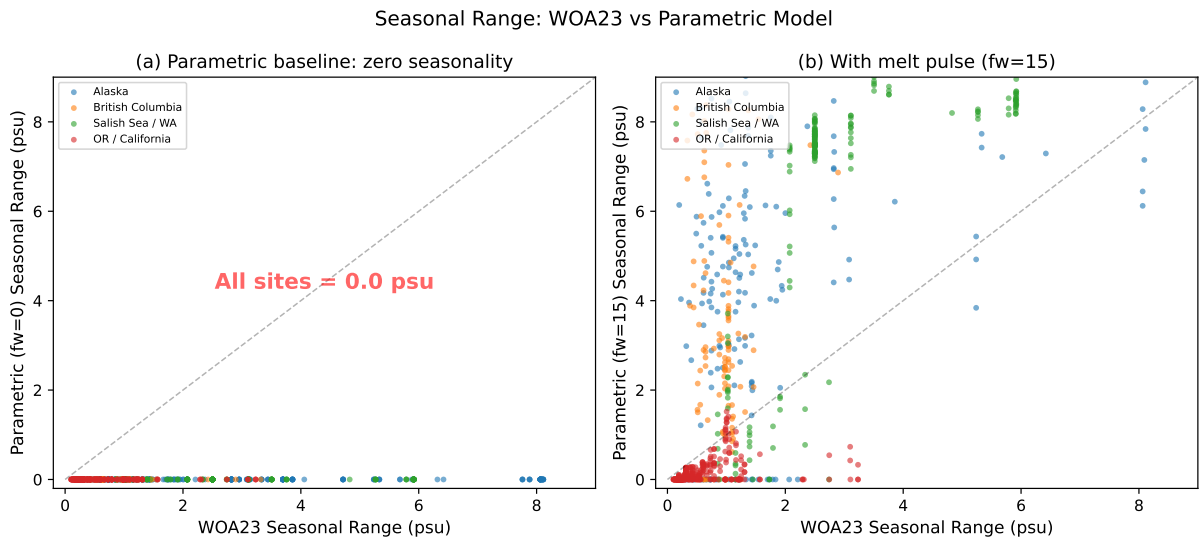

Figure 5: Seasonal range comparison. (a) Parametric baseline (`fw_strength=0`) has zero seasonal range at all sites. (b) With melt pulse (`fw_strength=15`), Alaska sites gain seasonality but California/Oregon remain flat. WOA23 shows real seasonal cycles at all latitudes.

5 Regional Profiles

Figure 6 presents mean seasonal profiles for each region group, including the WOA23 10th–90th percentile envelope, the parametric baseline (dashed), and the parametric model with melt pulse (`fw_strength=15`, solid grey). DFO lighthouse observations appear as black circles in the BC panel.

Key regional observations:

- **Alaska ($n = 275$):** Strong spring–summer freshening (May–August), particularly in fjord regions (AK-FN: seasonal range 3.14 psu, AK-PWS: 6.86 psu). The WOA23 10th–90th percentile shading spans 4–6 psu, showing high within-group heterogeneity. The parametric baseline sits near the upper bound of the WOA23 envelope. The melt pulse (fw=15) overshoots the WOA23 mean in summer because the cosine pulse applies the same timing (June 15) and amplitude scaling to all Alaska sites, while WOA23 shows earlier freshening in some subregions.

- **British Columbia ($n = 111$):** Moderate seasonality ($\sim$1.0 psu mean range). The parametric baseline aligns well with the WOA23 mean year-round. DFO lighthouse data generally fall within the WOA23 envelope, validating both datasets in this region.

- **Salish Sea / WA ($n = 182$):** The largest mismatch. WOA23 mean salinity is $\sim$28.4 psu (near the critical disease threshold), while the parametric model predicts $\sim$31.2 psu. The parametric model misses both the low mean and the 2.9 psu seasonal swing. Summer freshening drives WOA23 values to 26–27 psu in June–July, well into the suppression zone.

- **OR / California ($n = 328$):** WOA23 sits at $\sim$33.1 psu year-round with modest winter freshening (February minimum, range 0.58 psu). The parametric model runs $\sim$2.5 psu too low. Despite the small seasonal range, the consistent bias means the model systematically under-predicts the full-marine conditions that favour *Vibrio* growth.

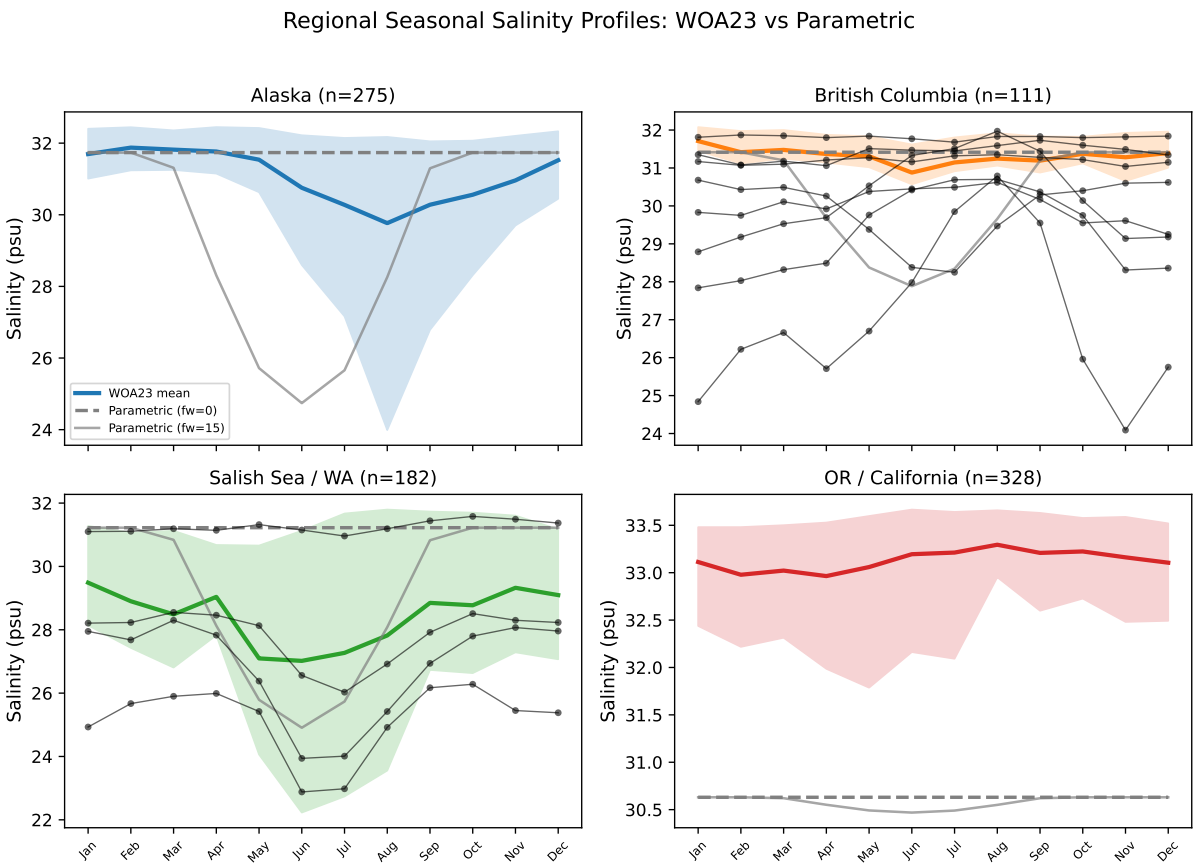

Figure 6: Regional seasonal salinity profiles. Coloured line: WOA23 mean; shading: 10th–90th percentile range. Dashed grey: parametric baseline (fw=0). Solid grey: parametric with melt pulse (fw=15). Black circles: DFO lighthouse observations (visible in BC panels).

6 Data Quality

Before drawing conclusions about the parametric model's performance, we must assess the quality of the WOA23 benchmark itself. Two concerns are paramount: the high fraction of sites requiring nearest-neighbour fill, and the accuracy of WOA23 against independent in-situ observations.

6.1 Nearest-Neighbour Fill Assessment

Of the 896 sites, only 235 (26.2%) fall within a WOA23 ocean grid cell. The remaining 661 (73.8%) required nearest-ocean-cell assignment. This high proportion reflects the coastal and estuarine nature of the SSWD-EvoEpi network — rocky shores, fjord heads, and convoluted coastlines that lie inside the 0.25° WOA23 land mask.

Figure 7 maps the extraction method for each site. The maximum NN distance is 0.177° (≈17 km), with a mean of 0.094° (≈9 km). Most NN-filled sites are within one grid cell of the coast. Sites with the largest NN distances are deep-fjord locations where the nearest ocean cell represents open-coast conditions rather than true fjord-head salinity.

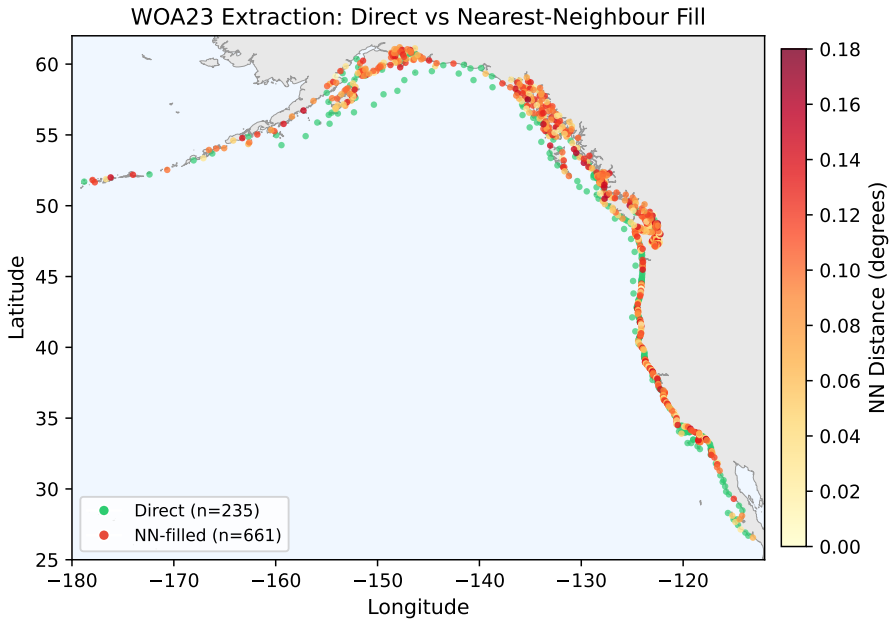

Figure 7: WOA23 extraction method. Green: direct ocean cell (235 sites). Yellow–red gradient: NN-filled (661 sites), coloured by distance to nearest ocean cell (max 0.177°).

6.2 DFO Lighthouse Validation

The 12 DFO lighthouse stations provide independent ground truth. Figure 8 shows monthly profiles for each station, comparing DFO observations (black), WOA23 (blue), and parametric (red). Overall, WOA23 achieves a mean RMSE of 1.58 psu against DFO, compared to 2.26 psu for the parametric model. The comparison is nuanced:

- **Both models perform well:** Langara Island (WOA23 RMSE 0.28, parametric 0.27), Pine Island (0.25 vs 0.20), Bonilla Island (0.41 vs 0.33). These are exposed outer-coast stations with salinities near the parametric formula's calibration point.
- **WOA23 clearly wins:** Chrome Island (WOA23 0.72 vs parametric 3.55), Departure Bay (1.51 vs 6.21), Entrance Island (1.26 vs 4.67). These Strait of Georgia stations have true salinities 4–6 psu below the parametric prediction. WOA23 captures the estuarine influence; the parametric formula cannot.
- **Both struggle:** Nootka Point (WOA23 5.02, parametric 4.73). This highly variable site with seasonal range >7 psu is poorly captured by both the 0.25° grid and the parametric formula.
- **Parametric wins:** Race Rocks (WOA23 2.67, parametric 0.18). Strong tidal mixing at this site produces salinities close to the open-ocean value captured by the formula, while WOA23's nearest ocean cell may be offset.

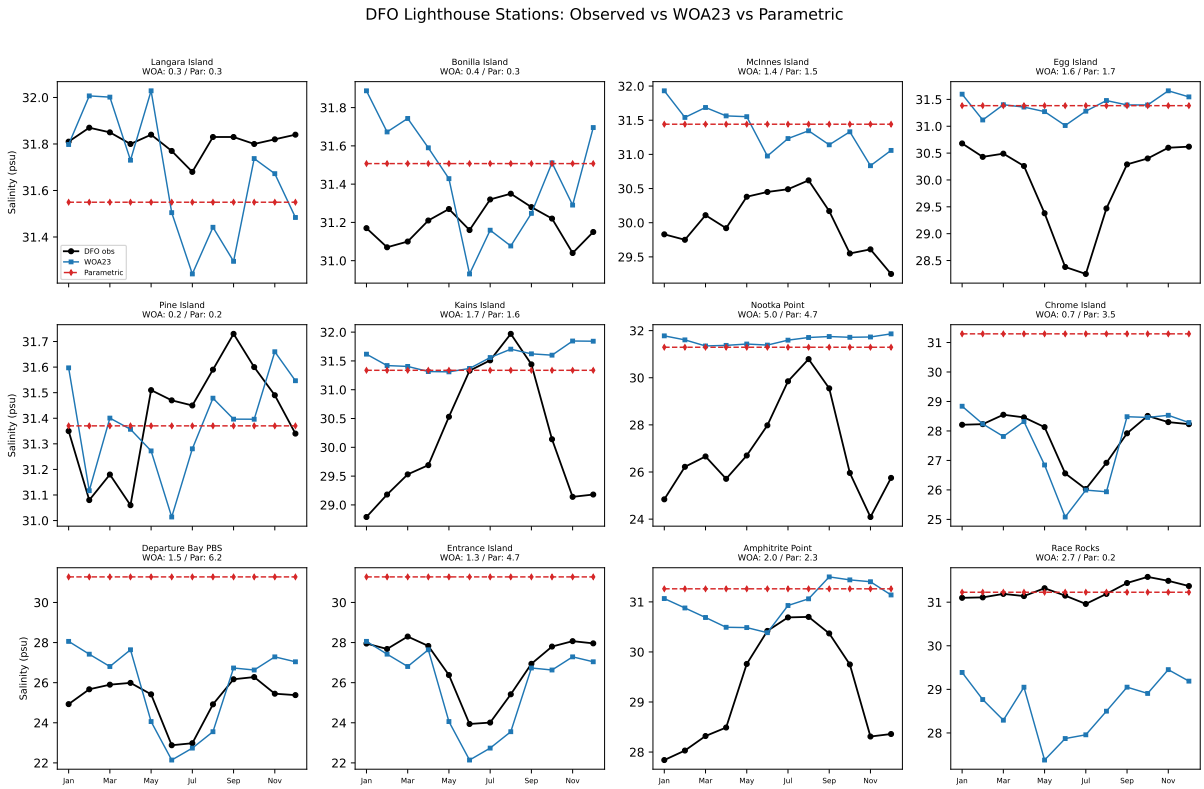

Figure 8: Monthly salinity at 12 DFO lighthouse stations. Black circles: DFO observed climatology. Blue squares: WOA23. Red diamonds (dashed): parametric model. RMSE shown in each panel title (WOA23 / parametric). Mean RMSE: WOA23 = 1.58, parametric = 2.26 psu.

7 Implications for Disease Modelling

The preceding sections establish that the parametric model has large spatial biases and no seasonal cycle. But do these errors actually matter for disease predictions? This section translates salinity errors into disease model outputs using the nonlinear salinity modifier (Equation 2).

7.1 Suppression Comparison

Figure 9 presents the most consequential comparison: June disease transmission suppression under the parametric baseline versus WOA23-informed salinity.

Under the **parametric baseline**, all 896 sites have June salinity > 30 psu, producing a salinity modifier of 1.0 everywhere. Zero sites experience any suppression. The parametric model is entirely blind to salinity-mediated disease dynamics.

Under **WOA23**, 143 sites show $> 1\%$ transmission suppression. The Salish Sea and Washington sites are most affected: mean suppression of 16.0% (maximum 54.5%), because WOA23 June salinity in this region averages 27.02 psu — below the 28 psu threshold. This represents a qualitative difference in model behaviour: the parametric model predicts full transmission everywhere, while WOA23 reveals a substantial suppression zone in the most ecologically sensitive region of the network.

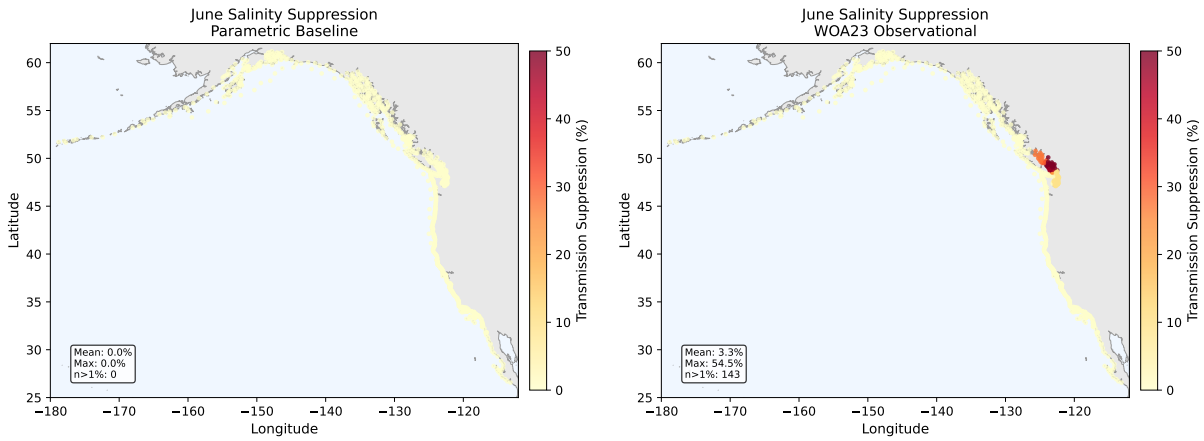

Figure 9: June transmission suppression (%). Left: parametric baseline (0% everywhere — all sites above 30 psu). Right: WOA23 (up to 54.5% suppression in the Salish Sea). The parametric model misses the entire suppression signal.

Table 2 quantifies the regional suppression differences. The Salish Sea is the critical region: the parametric model places it at 31.22 psu (full transmission), while WOA23 shows 27.02 psu (16% mean suppression). This 4.2 psu June bias directly translates to a qualitative error in disease predictions for this region.

Table 2: June salinity and disease suppression by region group.

Region	June Salinity (psu)		Suppression (%)	
	Parametric	WOA23	Parametric	WOA23
Alaska	31.74	30.75	0.0	0.0
British Columbia	31.42	30.88	0.0	0.3
Salish Sea / WA	31.22	27.02	0.0	16.0
OR / California	30.63	33.19	0.0	0.0

7.2 Latitude Gradient and the AK–CA Asymmetry

Figure 10 shows the latitude–salinity relationship in June. Panel (a) reveals the real WOA23 pattern: June salinity is *negatively* correlated with latitude ($r = -0.46$), meaning high-latitude sites tend to be *fresher* in June due to snowmelt and glacial discharge. This is the opposite of what one might naively expect from a simple "colder = saltier" assumption.

Panel (b) shows the parametric model: a perfect positive linear relationship ($r = 1.0$) by construction, with all values compressed between 30.05 and 31.92 psu. The formula assumes higher-latitude sites are slightly saltier, which is approximately true for annual means in open-ocean settings, but becomes wrong in June when freshwater forcing is strongest.

Panel (c) highlights the resulting bias pattern: the parametric model overestimates June salinity at high latitudes (Alaska) and underestimates at low latitudes (California). The two errors partially cancel in the global mean bias, masking the regional pattern.

This asymmetry matters for the AK–CA disease comparison. The parametric model predicts nearly identical salinity (and therefore transmission) across the entire domain in June. WOA23 shows that Alaska sites are substantially fresher, potentially experiencing salinity-mediated sup-

pression that the parametric model ignores, while California sites are substantially saltier, experiencing full transmission that the parametric model also misrepresents.

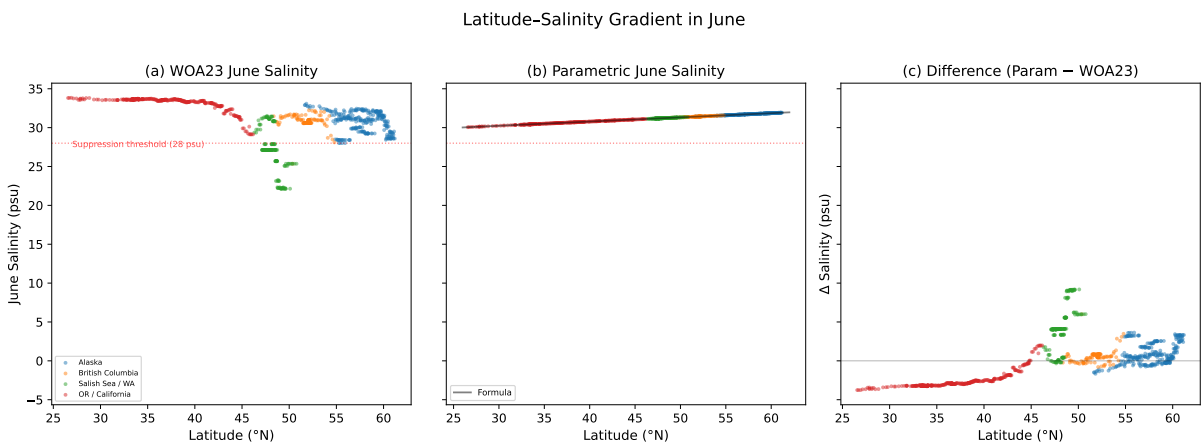

Figure 10: Latitude–salinity gradient in June. (a) WOA23: negative correlation ($r = -0.46$), higher latitudes fresher due to melt. (b) Parametric: perfect positive linear ($r = 1.0$) by construction. (c) Bias (parametric − WOA23): systematic overestimation in Alaska, underestimation in California. Red dashed line at 28 psu indicates the disease suppression threshold.

8 Recommendation

Based on this analysis, we recommend **replacing the parametric baseline with WOA23 monthly climatology** as the default salinity input for the SSWD-EvoEpi model. This change would:

1. **Introduce site-specific seasonal cycles.** Mean seasonal range of 1.68 psu, capturing real freshwater forcing that the constant baseline ignores entirely.

2. **Correct systematic regional biases.** The +2.8 to +4.8 psu overestimation in the Salish Sea — where disease dynamics are most sensitive to salinity — would be eliminated. The −2.5 to −3.4 psu underestimation along California would also be corrected.

3. **Enable salinity-mediated suppression.** The 143 sites with $> 1\%$ June suppression would gain this signal, particularly the 182 Salish Sea/WA sites that currently show zero suppression.

4. **Capture correct seasonal timing.** Winter-storm freshening in California (February minimum) and late-glacial melt in Alaska (August minimum) would be represented, rather than a universal June 15 pulse.

5. **Reduce overall RMSE.** From 2.42 psu (parametric vs WOA23) to ∼1.58 psu (WOA23 intrinsic uncertainty vs DFO ground truth).

8.1 Implementation Notes

The WOA23 monthly climatology for all 896 sites is already extracted and stored as a (896×12) array. Integration requires:

- Replacing the `ocean_baseline(lat)` call with a lookup into the WOA23 monthly array for the current site and month.

- The existing `fw_strength` mechanism could be applied *on top of* the WOA23 baseline for sites where additional sub-grid freshwater forcing (fjord heads, river mouths) is needed, though the base WOA23 cycle already captures much of this variability.
- The WOA23 data file adds <100 KB to the model's data requirements.

8.2 Caveats

WOA23 is not without limitations:

- **Resolution:** The 0.25° grid smooths over sub-grid features. Fjord-head sites may have surface salinity considerably lower than the nearest WOA23 ocean cell.
- **NN fill uncertainty:** 73.8% of sites use NN-filled values. For deep-fjord sites, true salinity is likely lower than WOA23 due to unresolved river and glacial inputs.
- **Interannual variability:** WOA23 provides a long-term climatological mean, not year-specific conditions. Years with anomalous precipitation or snowmelt may deviate substantially.
- **Coastal accuracy:** The DFO validation shows RMSE of 1.58 psu against in-situ data, with individual stations reaching 5.02 psu (Nootka Point). WOA23 is an improvement, not ground truth.

Despite these limitations, WOA23 climatology represents a substantial improvement over the parametric formula for all three dimensions of salinity representation: spatial heterogeneity, seasonal dynamics, and absolute accuracy.

References

1. Reagan, J. R., et al. (2024). World Ocean Atlas 2023. NOAA National Centers for Environmental Information.
2. Fisheries and Oceans Canada. BC Lighthouse Sea-surface Temperature and Salinity Data.
3. Eisenlord, M. E., et al. (2016). Ochre star mortality during the 2014 wasting disease epizootic: role of population size structure and temperature. *Philosophical Transactions of the Royal Society B*, 371(1689), 20150212.